

Lecture 0 — Course Overview

COSE212: Programming Languages

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Basic Information

Instructor: Minseok Jeon

- **Position:** Assistant professor in EECS, DGIST
- **Expertise:** Programming Languages
- **Office:** 211 E7 Bldg.
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- **Office Hours:** 9:00am–11:59am Mondays (by appointment)

TAs:

- Sanggyu Oh (sg549@dgist.ac.kr)

About This Course

This is not an introductory course on programming.

- You will not learn particular programming languages



- You will not learn how to write programs in those languages

About This Course

Instead, in this course you will learn

- how programming languages are designed and implemented
- fundamental principles of modern programming languages
- thinking formally and rigorously

To succeed in this course, you must

- have basic programming skills
- be familiar with at least two PLs (e.g., C, Java)
- have taken Theory of Computation, Discrete Math, etc
- **be prepared to learn new things**

Design and Implementation of Programming Languages

You will learn programming language concepts by designing and implementing our own programming language system.

- We will define a programming language. For example, “factorial” is written in our language as follows:

```
let x = read in
letrec fact(n) =
  if iszero n then 1
  else ((fact (n-1)) * n)
in (fact x)
```

- We will design and implement an interpreter for the language:

Program $\rightarrow$ Interpreter $\rightarrow$ Result

- We will design and implement a type checker for the language:

Program $\rightarrow$ Type Checker $\rightarrow$ Safe/Unsafe

Desirable Achievement

By the end of this course, I hope you are able to:

- **Develop a programming language to address your own problems**
 - Design a language to solve a specific (i.e., your own) problem
- Programming languages are not just tools to use – they are tools you can **create** to solve problems more effectively

Functional Programming

The secondary goal of this course is to be familiarized with functional programming, which encourages using pure functions rather than making side effects.

- Functional programming is one of the major programming paradigms adopted in modern programming languages such as Python, JavaScript, C++, Java8, Scala, Go, etc.
- In this course, you will learn functional programming with OCaml¹ and use it to implement programming languages.

¹<https://ocaml.org>

Topics

- **Part 1 (Preliminaries):** inductive definition, basics of functional programming, recursive and higher-order programming
- **Part 2 (Basic concepts):** syntax, semantics, naming, binding, scoping, environment, interpreters, states, side-effects, store, reference, mutable variables, parameter passing
- **Part 3 (Advanced concepts):** type system, typing rules, type checking, soundness/completeness, automatic type inference, polymorphic type system, lambda calculus, program synthesis

Course Materials

- Self-contained slides will be provided.
 - You are required to attend every class (otherwise, it'd be difficult to catch up)
- Hakjoo Oh. Introduction to Principles of Programming Languages. (pdf will be provided)
- (Supplementary) Essentials of Programming Languages (Third Edition) by Daniel P. Friedman and Mitchell Wand. MIT Press.

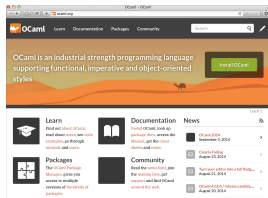


Grading

- Attendance & Participation – 10%
- Homework assignments – 10%
- Midterm – 40%
- Final – 40%

Programming in ML

- ML is a general-purpose programming language, reflecting the core research achievements in the field of programming languages.
 - higher-order functions
 - static typing and automatic type inference
 - parametric polymorphism
 - algebraic data types and pattern matching
 - automatic garbage collection
- ML inspired the design of modern programming languages.
 - C#, F#, Scala, Java, JavaScript, Haskell, Rust, etc
- We use OCaml, a French dialect of ML:



<http://ocaml.org>

Questions?

(!) Special Project

Project Goal: Design a programming language that is AI-resistant but human-friendly.

Challenge:

- Existing AI systems (ChatGPT, Claude, etc.) can easily solve programming assignments in conventional languages
- This undermines the learning process and academic integrity

Your Task:

- Design a new programming language where:
 - AI models struggle to write correct solutions
 - Human programmers can still solve assignments effectively
- Consider: syntax design, semantic features, unconventional paradigms
- Deliverable: Language specification + justification of design choices

This is an open-ended research project – be creative!

(!) Special Project

Reward:

- Students who successfully design such a language will receive **the highest score**

How to Participate:

- Email the instructor (minseok_jeon@dgist.ac.kr) if you want to challenge this project

Good luck!